
Does the Source of Carrier Image Affect Steganographic Detectability?

Final Presentation

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Agenda

1. Motivation recap
2. Research questions
3. Experimental design and methods
4. Statistical analysis
5. Results across RQ1–RQ5
6. Limitations and contribution

Motivation Recap

- Steganalysis detectability depends on carrier-image statistics, not only on embedding logic (Petitcolas et al., 1999; Fridrich & Kodovsky, 2012).
- Photographic carriers dominate benchmarks, yet diffusion outputs are now everywhere and carry distinct statistical traces (Corvi et al., 2023).
- Closest prior work studies AI carriers under adaptive embedding and reports P_E , not AUC (Méreux et al., 2024).

Core question

Do detectors validated on photographs remain equally effective when the carrier is ML-generated, under identical embedding settings?

Research Questions

RQ1 (Primary, confirmatory)

Does carrier source (real vs. pooled ML) affect detectability under matched embedding settings?

RQ2 (Primary, confirmatory)

Within ML carriers, does the choice of generator (SDXL vs. FLUX) change detectability?

RQ3 (Exploratory)

Does payload size change the detectability gap between real and ML carriers?

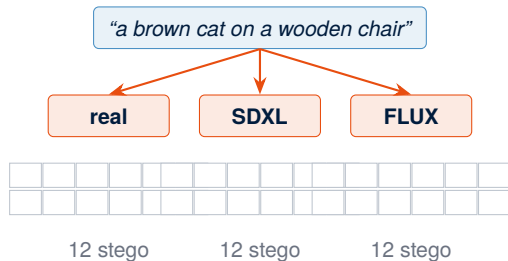
RQ4 (Exploratory)

Do the spatial (LSB+PNG) and frequency (DCT-LSB+JPEG) branches show different carrier-source effects?

RQ5 (Verification)

Does AES-256-CBC payload encryption affect detectability? Expected null result.

Caption-Matched Carriers, $3 \times 2 \times 3 \times 2$ Factorial



Per-group recipe

- Three caption-matched covers per group
- $2 \text{ methods} \times 3 \text{ payloads} \times 2 \text{ encryption} = \mathbf{12 \text{ stego per source}}$
- Per group: **3 cover + 36 stego** = 39 images

Datasets (1,000 groups)

- Real: COCO + Flickr30k captions
- ML-A: SDXL 1.0; ML-B: FLUX.1-schnell
- Grayscale 512×512 , dual storage: PNG (spatial) + JPEG Q=95 (frequency)

Embedding Methods

Spatial: LSB substitution

Replace the least-significant bit of each pixel, $k = 1$, row-major. Lossless PNG storage required.

cover pixel



bit=1



stego pixel

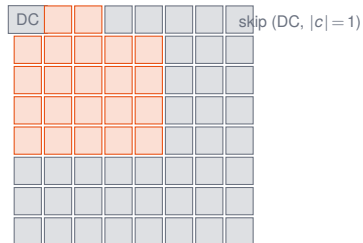


One pixel – one payload bit. Detected by RS, χ^2 , Sample Pairs.

Frequency: DCT-LSB (JSteg-style)

Flip the LSB of *non-zero* quantised AC coefficients in each 8×8 block. Skip DC and $|c| = 1$ (Westfield-pair stable). JPEG Q=95.

used (non-zero AC)

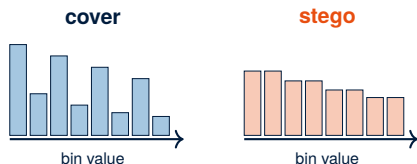


Detected by χ^2 -DCT and Calibration χ^2 .

Detector Mechanisms

Pairs-of-Values intuition

χ^2 tests the prediction that LSB embedding equalises the counts of value pairs $\{2k, 2k+1\}$. Bars within each pair converge in stego images.



χ^2 score rises whenever paired bars are mismatched in the cover and equalise in the stego.

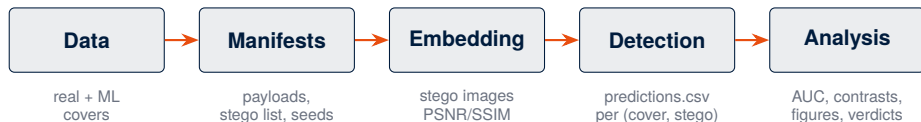
Spatial

- **RS** (Fridrich 2001): regular/singular pixel-group ratios
- χ^2 (Westfeld 1999): PoV (left)
- **Sample Pairs** (Dumitrescu 2003): traces on adjacent-pixel multisets

Frequency

- χ^2 -**DCT** (Westfeld 1999): PoV on AC coefficients
- **Calibration** χ^2 (Fridrich 2003): non-block-aligned crop + recompress reference

Pipeline Structure



Five sequential stages

- Data and manifests are reproducible from seeds.
- Embedding writes one stego file per condition.
- Detection scores cover+stego pairs.
- Aggregation derives all RQ outputs.

Run isolation

- Each run lives in `runs/<id>/`.
- Config frozen in `config.json`.
- Re-runs are idempotent: done work is skipped.

Web Platform and Live Monitoring

Architecture

- Local Python HTTP server (stdlib only)
- Single-page browser client, no build step
- Endpoints: list / launch / preview / kill / detail
- Launch spawns the runner as a subprocess

Live observability

- Subprocess stdout via Server-Sent Events
- Browser terminal panel shows progress live
- Preview validates config before any compute
- Run-detail surfaces metrics, figures, gallery

Power estimate in the drawer: Hanley–McNeil computed live in JavaScript, colour-coded against the proposal threshold $\delta = 0.05$.

Statistical Analysis

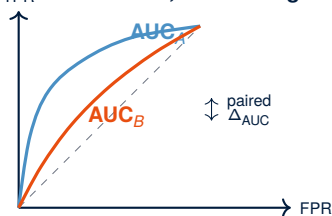
Tests

- DeLong paired ROC test for correlated AUCs (DeLong 1988)
- Bonferroni–Holm correction across the confirmatory family
- 95% CIs for exploratory comparisons (RQ3, RQ4)
- Equivalence margin ± 0.025 AUC for RQ5

Power (Hanley–McNeil)

- $\alpha = 0.05$, target power 0.80
- Confirmatory family size: 15 strata per RQ
- Min detectable δ_{AUC} at $N=1000$: RQ1 ≈ 0.040 , RQ2 ≈ 0.047

DeLong: same cover set, two scorings



Each Δ_{AUC} comes from a single stratum; Holm controls the family-wise error rate across 15 strata.

RQ1 – Real vs. Pooled ML

Test

Paired DeLong per stratum (detector $\times$ method $\times$ payload), Holm-adjusted across 15 strata.

Headline finding

- Pooled Δ_{AUC} : [*to fill from exp1_*_contrasts.csv*]
- Significant strata (Holm $p < 0.05$): [$n / 15$]
- Verdict: [*supported / mixed / not supported*]

Interpretation

- Direction of the effect, consistent across detectors or not.
- Where is the gap strongest – spatial or frequency branch?
- Cite exp1_real_vs_pooled_ml.png forest.

RQ2 – SDXL vs. FLUX within ML

Test

Paired DeLong, same 15-stratum family, Holm-adjusted. Caption-matched cover pairs across the two generators.

Headline finding

- Pooled Δ_{AUC} : [*from* exp2_*_contrasts.csv]
- Significant strata: [*n / 15*]
- Verdict: [*supported / mixed / not supported*]

Power gain at N=1000

At N=1000 the minimum detectable δ for RQ2 is ≈ 0.047 , below the proposal threshold. RQ2 is now adequately powered.

RQ3 and RQ4 – Exploratory

RQ3: Payload interaction

- Real-vs-ML gap by payload level (low / med / high).
- Trend: [*increasing* / *non-monotonic* / *flat*]
- Cite `exp3a_payload_level_auc.png` and the `source`×`payload` heatmap.

RQ4: Branch interaction

- Spatial source gap minus frequency source gap, pooled per branch.
- Pooled $\Delta\Delta$: [*from* `exp4_*_contrasts.csv`]
- Cite `exp4_spatial_vs_frequency.png`.

Reporting convention

RQ3 and RQ4 are exploratory: we report point estimates and 95% CIs, no Holm correction, no significance verdict.

RQ5 – Encryption Invariance (verification)

Test

Paired DeLong between plain and AES-256-CBC stego on shared cover groups. Equivalence margin ± 0.025 AUC per stratum.

Headline finding

- Strata within margin: $[n / 45]$
- Pooled Δ_{AUC} : [*near zero, expected*]
- Verdict: encryption invariant / not supported

Why the null matters

A non-null result would indicate the detectors are reacting to payload structure rather than the embedding distortion itself. The null is the proposal-aligned outcome.

Limitations

Acknowledged deviations

- ml_b: PixArt- α replaced with FLUX.1-schnell (PixArt unavailable on HF Inference API).
- Two DCT-branch detectors vs. three spatial; branch contrasts pool over detectors.

Scope-bounded

- Two generator families only – cannot disentangle architecture from training data in RQ2.
- Grayscale only; no colour information.
- Classical detectors only; no learned baselines.

Contribution and Minimum Deliverable

What this study contributes

- Caption-matched dataset across three carrier sources under identical embedding.
- Training-free AUC-based comparison with DeLong and Holm correction.
- Pre-computed RQ verdicts and power analysis as reproducible artefacts.

Minimum deliverable met

- One encryption scheme (AES-256-CBC). ✓
- Two embedding approaches: spatial LSB and frequency DCT-LSB. ✓
- Defensible answers to RQ1 and RQ2. ✓

Thank you

Questions and Discussion

Code and run artefacts: `runs/<id>/`
